

RESEARCH ARTICLE

Raman detection system for in situ oxidation and ablation analysis in hypersonic wind tunnel

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Abstract

We have developed a system for remote Raman spectra detection in hypersonic wind tunnel, which can be used in extreme environments such as ultra-high temperature, high pressure, and complex airflow to measure in situ Raman spectra of ceramic matrix composites during oxidation, phase transformation or ablation. The designed system avoids stray light generated by windows and airflow through oblique incidence of excitation light. The time-resolved method combined with pulsed laser and intensified charge-coupled device (ICCD) was used to reduce the thermal emission background. An optical structure design based on fiber bundle is adopted to suppress the influence of aero-optical effects and mechanical vibration on Raman spectrum acquisition. We demonstrated the feasibility of in situ Raman spectra detection in a Mach 5 combustion wind tunnel system. The design of anti-interference prevents Raman spectra intensity from the influence of wind tunnel testing conditions.

KEYWORDS

ablation, Raman spectroscopy, system design, ultra-high temperature composites, wind tunnel

1 | INTRODUCTION

The ultra-high temperature composites are particularly attractive as high-temperature structural element for aerospace equipment, such as in re-entry heat shields, rocket nozzles, aeronautic jet engines, turbo pump rotors, and leading edges.^[1,2] The composites can be used in extreme environments with high heat fluxes, high enthalpy, and high pressure. In this type of environment, operation temperatures may increase rapidly from room temperature to over 3273 K and last from several seconds to several hundred seconds. Oxidation and phase transition of the composites are significant issues. Thus, investigations of the oxidation mechanisms and thermochemical ablation of composites are of considerable importance.^[3] The main experimental methods used to test these composites include oxyacetylene flame ablation,^[4,5] plasma arc-jet ablation,^[6,7] and wind tunnel

tests.^[8] Among these methods, the wind tunnel tests can best simulate the service environment.^[3,9,10]

The oxidation mechanism of composites has been studied in recent decades; the oxide layer formed on the surface during oxidation is known to be a key point in the oxidation resistance of composites. A more viscous and more stable glassy oxide layer is able to decrease the diffusion of oxygen at higher temperatures.^[11] Therefore, it is very important to study the behavior of oxide layer at high temperature to analyze the oxidation mechanism of composites. However, in most cases, properties of the oxide layer are inferred from post-test analyses, such as scanning electron microscope (SEM), X-ray diffraction (XRD), and energy dispersive spectroscopy (EDS).^[12,13] The distribution information of oxidation products was obtained by analyzing the oxidation cross section of samples. A few studies attempted to investigate the in situ, and thus continuous behavior during a complete

oxidation test. The thermogravimetric analysis (TGA) is a common method to study the oxidation process,^[14] but mass variations as a function of time only give insights to the sum of the mass gains and mass losses. In some other studies, silicon emissions were detected by optical emission spectroscopy,^[15] to measure the erosion products from the actively oxidized samples. However, the elements have to be in an excited state, and it cannot be applied in situation of passive oxidation. A better understanding of the oxidation process, such as the formation and content of oxide layer and the stress of ceramic coating on the surface can provide more information for further explaining the oxidation mechanism of composites.

Raman spectroscopy is the informative method for studying oxidation, phase transformations, and a behavior of the structure of composites. It is widely adopted to determine the composition of oxidation products^[16,17] and the residual stress of bulk coatings.^[18,19]

In addition to the offline detection after high temperature treatments, in situ Raman spectroscopy allows monitoring the oxidation kinetics while providing time-resolved information about structural changes during the treatment. Although real-time observation/processing methods have been developed and used in many scenes,^[20–24] in situ measurements of Raman spectra in wind tunnels have not been reported. Under the action of hypersonic flow, the ablation process of high temperature composites is the result of physical ablation, chemical ablation, and mechanical stripping. The information of

the process when the materials were chemically reacting and physically evolving is the critical parameters to fully reveal the ablation mechanism. Therefore, in situ Raman spectroscopy detection in wind tunnel is of great significance.

In this work, we designed and constructed a Raman spectroscopy system and successfully obtained in situ Raman spectra of oxides of composites in a combustion wind tunnel at Mach 5.

2 | SYSTEM CONSTRUCTION

Raman system was set up on an air floating vibration isolation platform on the side of the wind tunnel. Figure 1 illustrates the optical setup and optical paths. The system consists of two main parts: The excitation part and the Raman detection part. Besides, temperature at the sample was monitored via a thermal infrared imager (Lumasense MCS640).

2.1 | Excitation

Raman scattering was excited by a compact Q-switched Nd: YAG laser (Innolas Spitlight DPSS250) operating at 532 nm having 100-Hz repetition rate, 8-ns pulse duration, and a maximum energy of 120 mJ. The compact thermally stabilized monolithic structure of the Nd: YAG

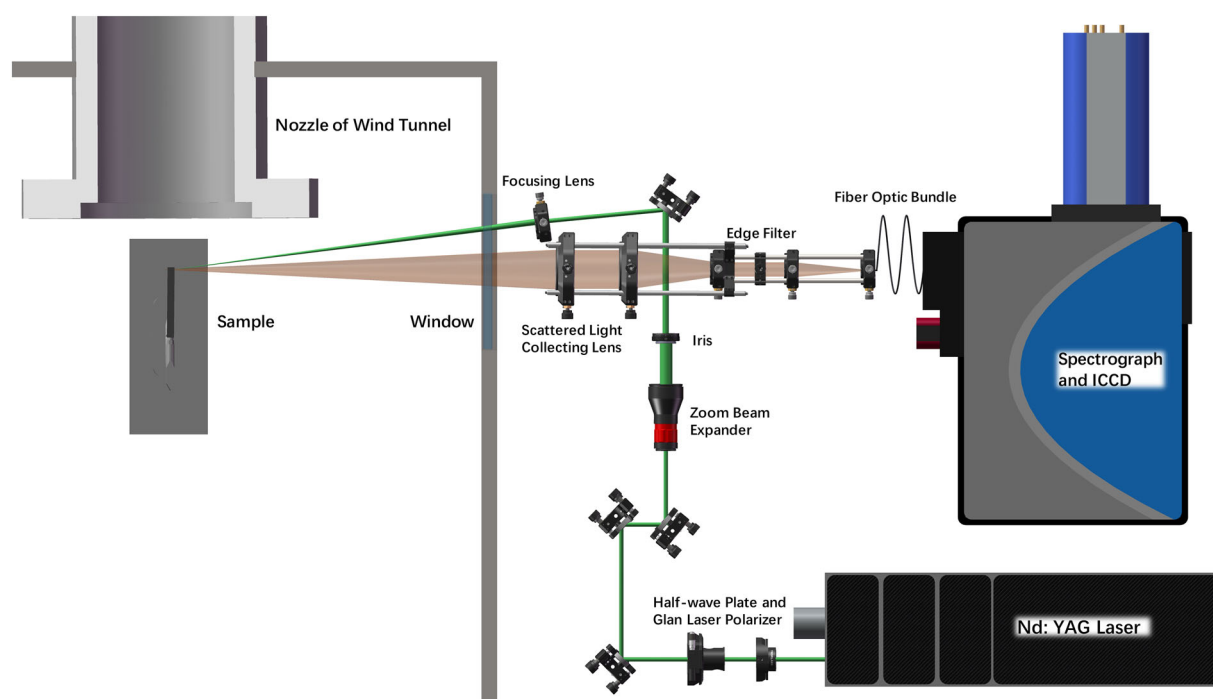


FIGURE 1 Schematic diagram of the designed system [Colour figure can be viewed at [wileyonlinelibrary.com](https://onlinelibrary.wiley.com/doi/10.1002/jrs.6449)]

laser provides excellent mechanical stability and superior beam geometry even under changing environmental conditions such as temperature variation and mechanical vibrations. The output laser energy can be controlled by a half-wave plate placed before the Glan laser polarizer. The height and beam diameter of the excitation light was adjusted by series of mirrors and a beam expander. An iris is used to make the edge of the laser spot clear and the energy distribution more uniform. Finally, the excitation light is focused on the sample through a lens. The size of the excitation light spot can be changed by adjusting the lens.

Most commercial Raman spectrometers utilize back-scattered geometry; this geometry results in intense Raman spectra.^[25] However, it is not suitable for detection in a wind-driven environment, because it has the disadvantage of being difficult to remove the noise imposed by back reflected excitation light from the window of the wind tunnel. More importantly, Raman signals generated by the excitation of the window or the airflow will also be collected and influence the measurement results; for example, silicon dioxide from the optical elements such as lenses and the window will affect the measurement of the oxidation process of silicon carbide.

The optical path of the Raman system designed in this work adopts the structure of oblique incident excitation light, as shown in Figure 2. Because the excitation light is separated from the collection light path, the Raman scattering generated by the additional components, such as windows or airflow, will not enter the collection light path, and only the Raman signal from the measurement point on the sample can be collected. Meanwhile, in order to enhance the collecting efficiency

of Raman as much as possible, without interfering with the detection signal collection, the excitation light is incident at the smallest possible angle. In this work, the excitation light was incident on the sample at an incident angle of 15° .

2.2 | Raman detection

The Raman signal is collected in the direction perpendicular to the window of the wind tunnel. The scattering light from the surface of the sample in wind tunnel was collected using a 2 in. achromatic doublet of focal length 500 mm. An edge filter (Semrock LP03-532RE) was set to subdue the Rayleigh scattering light before feeding into the fiber optic bundle. The linear end of the bundle cable was plugged into the spectrograph (Andor SR303I). The fiber array was aligned with the entrance slit.

The gain function of the ICCD (Andor iStar DH320T) was used to amplify the signal of the collected Raman spectra to improve the detection efficiency. The ICCD utilized in this work consisted of 2048×512 pixels, and the measurement range can cover from 100 to 2000 cm^{-1} with the grating of 1200 L/mm. This range allows coverage of the Raman spectrum of most high-temperature composites without rotating the grating, so that each excitation pulse can complete a fully signal acquisition. Therefore, Raman spectra can be obtained by excitation pulse accumulation with high efficiency to improve the SNR and shorten the spectrum acquisition time. Generally, materials with higher Raman activity require less pulse accumulation, such as zirconia or titanium oxide. They can usually complete a spectrum acquisition in 0.01

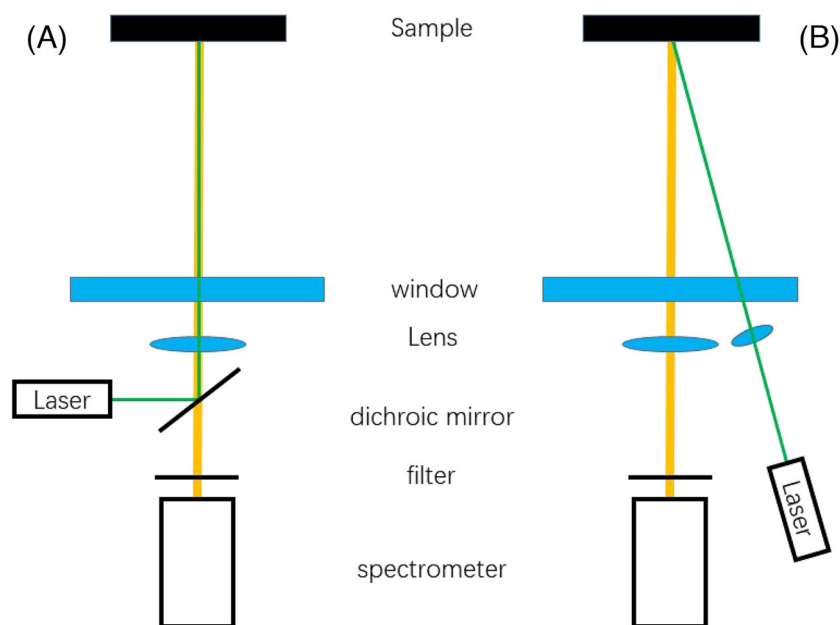


FIGURE 2 The geometry of the Raman excitation light: (A) backscattered; (B) oblique incident [Colour figure can be viewed at wileyonlinelibrary.com]

to 1 s. For materials such as silicon oxide, it takes 1 s or more to distinguish Raman peak from noise peak. The spectrometer and detector were calibrated using the 546.07 nm emission line from a mercury argon calibration lamp. Spectral resolution is about 15 cm^{-1} by reading the calibration light.

3 | OPTIMAL DESIGN

Different from the optical methods used in the laboratory, there are many interference factors when measuring Raman spectrum in situ in wind tunnel, for example, hypersonic turbulence and strenuous mechanical vibration. In this work, some optimization designs were adopted to implement in situ Raman measurement in a wind tunnel environment.

3.1 | Reducing the thermal emission

At high temperature, recording a Raman spectrum becomes increasingly difficult due to the thermal emission produced by hot samples. The method of time-resolved detection was widely used for probing materials at high temperatures by Raman spectroscopy.^[26] The basic idea here is to use the fact that Raman scattering is a very rapid phenomenon, whereas thermal emission is intrinsically continuous. Thus, the signal-to-noise ratio between the Raman scattering and the thermal radiation background can be enhanced by synchronizing the excitation laser pulse with the detection time of the corresponding Raman scattering signal through a timing controller. In this work, the laser with an 8 ns pulse width was synchronized by a timing controller with the ICCD, which provides a resolution of gate width time of 1 ns.

In order to determine the appropriate excitation pulse width, a series of detection experiments were conducted in the laboratory. We observed that Raman spectra can be measured in the range of $\pm 10\text{ ns}$ at the center of the laser pulse. Therefore, the performance of different gate widths of the ICCD from 4 to 20 ns was tested. Figure 3 is a schematic representation of the time evolution for time-gated Raman measurements in high temperature.

To determine the ICCD gate width time, the Raman signal strength was compared with the thermal radiation background. The Raman spectra of SiC samples with different gate widths at the same Raman peak at 1700 K are compared, as shown in Figure 4. The Raman peak at 790 cm^{-1} was selected.

The intensity of the collected thermal radiation background signal increases linearly with the gate width time,

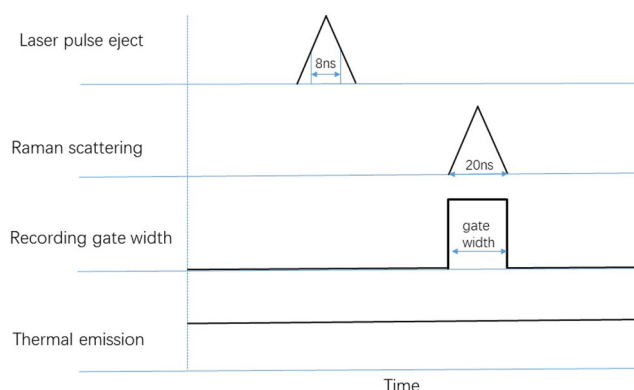


FIGURE 3 Schematic representation of time gating for Raman measurements [Colour figure can be viewed at [wileyonlinelibrary.com](https://onlinelibrary.wiley.com/doi/10.1002/jrs.6449)]

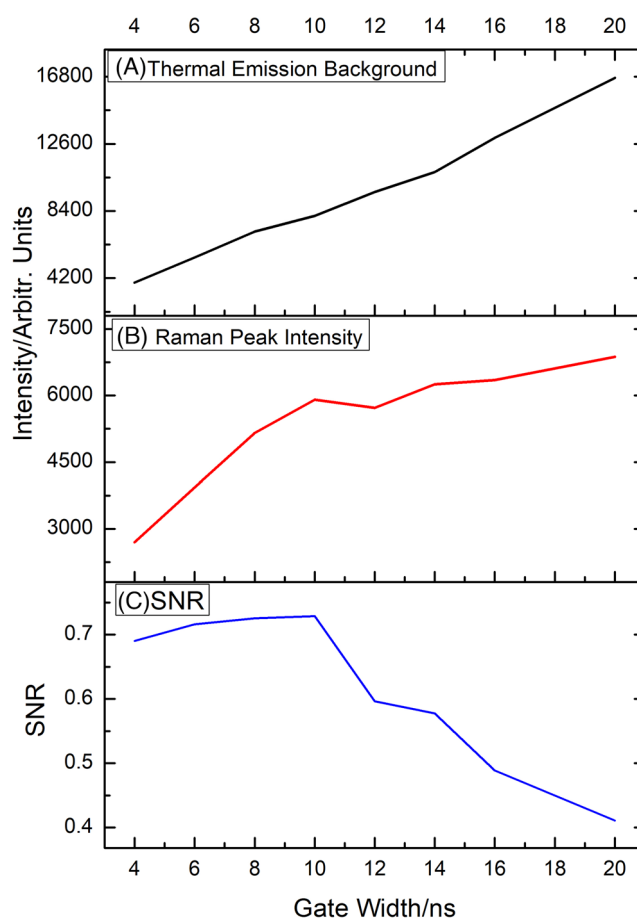


FIGURE 4 Signal collected at 1700 K with different gate width times: (A) The thermal emission background; (B) the intensity of Raman peak at 790 cm^{-1} of SiC; (C) the SNR of the Raman spectra [Colour figure can be viewed at [wileyonlinelibrary.com](https://onlinelibrary.wiley.com/doi/10.1002/jrs.6449)]

whereas the Raman signal intensity is mainly concentrated in the range of $\pm 5\text{ ns}$ (i.e., gate width of 10 ns) at the center of the laser pulse. The relationship between signal to noise ratio (SNR) and gate width is obtained by calculating the ratio of Raman intensity to thermal emission background.

It can be seen that the gate width of 10 ns is a better choice for 8 ns pulse laser as excitation source.

3.2 | Aero-optical distortion and mechanical vibration

The wind tunnel samples interact with the hypersonic air-flow to form a complex nonuniform flow field structure including a shock wave, expansion wave, and strong turbulent boundary layer.^[27–29] The influences of this nonuniform flow field on optical propagation and optical imaging are called aero-optical effects.^[30,31] The nonuniformity can induce a variation of the refractive index and thus lead to a deflection of the ray to the Raman system, which will introduce the signal intensity to decrease. Therefore, how to reduce the influence of the nonuniform airflow is a key topic for the optical measurement in wind tunnel.

Moreover, the inevitable mechanical vibration caused by wind tunnel is also a problem that cannot be ignored in precision optical detection. The vibration of the sample or optical path structure will cause the deviation of Raman measurement position. This also leads to errors in Raman spectral intensity, which affects the measurement and quantitative analysis of Raman spectra.

To solve this problem, a round-to-linear fiber optic bundle was utilized before the Raman scattering light enters the spectrometer. This fiber bundle contains multiple fibers arranged in a line configuration (linear) at one end and a circular configuration (round) at the other end. Raman scattering light was fed into the round end from the collection path and exited from the linear end to the slit of the spectrometer. The round end has an effective core diameter of 1 mm. By adjusting the optical path, the spot diameter of the collected Raman scattering light focused on the fiber bundle is 0.9 mm and located in the center, as shown in Figure 5. For the collection path of a

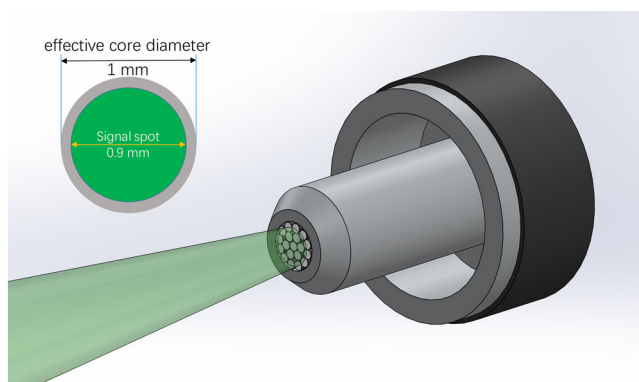


FIGURE 5 Schematic representation of the round end of the fiber optic bundle [Colour figure can be viewed at [wileyonlinelibrary.com](https://onlinelibrary.wiley.com/doi/10.1002/jrs.6489)]

Galileo telescope structure with a magnification of 0.4, the measurement spot on the sample can be allowed to have random jitter in the range of 0.125 mm. Estimating based on existing reports, the position deviation due to vibration and aero-optical effects is about 0.112 mm in our wind tunnel environment. It means that the Raman signal can be fully received by our design.

4 | WIND TUNNEL TEST

4.1 | Test conditions

In order to verify the feasibility of the designed Raman system, an in situ Raman test was performed in a combustion wind tunnel system, which could create a high-enthalpy, high Mach gas flow through the combustion of aviation kerosene. The specimen was mounted in a steel holder with water cooling. The test chamber was equipped with windows for optical diagnostics. In this work, the flow speed was set to 5 Mach. The total temperature was 2312 K.

4.2 | Materials

Zirconia is a very common component in thermal protection coatings and materials. Zirconia has a melting point of 2973 K and can withstand high temperature conditions in wind tunnel tests. It not only has high Raman activity but also has obvious phase transformation process at 1400 and 2600 K, which is suitable for in situ experiments in wind tunnels to test the detection ability of Raman system. Therefore, the C/C-ZrB₂-ZrO₂ composites were selected as test samples, and the reliability of the system was tested by continuous measurement of the in situ Raman spectra in wind tunnel experiments.

5 | RESULTS AND DISCUSSION

5.1 | In situ Raman spectra in wind tunnel

In the wind tunnel experiment, the heating time of the sample was 50 s, and the in situ Raman spectrum was completely recorded by the Raman system. By aligning the thermal imager data frame by frame with the laser pulse, the temperature of the sample point at the corresponding time of each Raman spectrum can be obtained.

The in situ Raman spectra measured during the wind tunnel test was shown in Figure 6. At room temperature, the peaks (178, 190, 334, 349, 382, 474, 502, 616,

640 cm^{-1}) can be clearly assigned to Raman active modes ($9A_g + 9B_g$) of monoclinic ZrO_2 . Below 1273 K , all phonon frequencies show a linear decrease and an increase in linewidth to follow with the increasing temperature. When the temperature increased from 1273 to 1473 K , the observed spectra change abruptly, which indicates a phase transition. Those lines originating from the

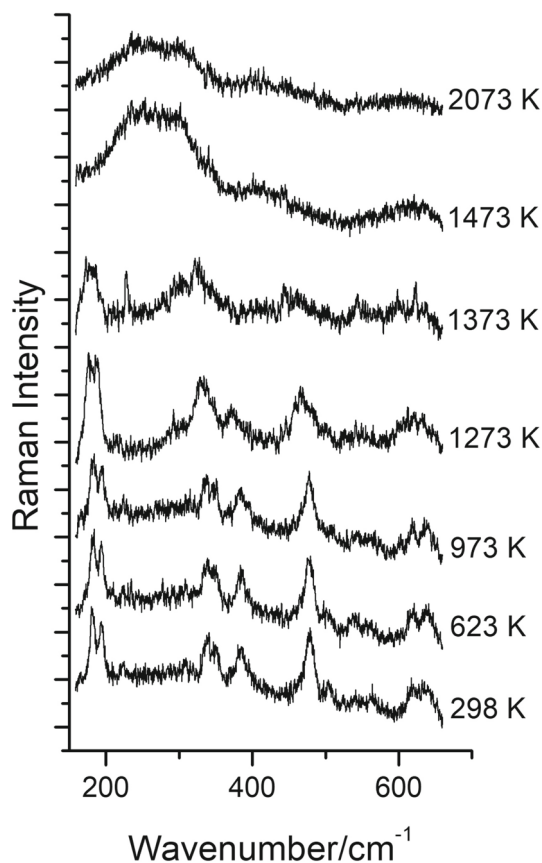


FIGURE 6 The in situ Raman spectra measured in the wind tunnel test

monoclinic phase become very weak, greatly shift, or disappear. New broad lines ($246, 295, 429, 621\text{ cm}^{-1}$) that represent the Raman active modes of tetragonal zirconia appear instead.

This process is approximately the same as reported by the previous work.^[26,32] Spectra obtained at the highest temperature 2073 K reached in this study show that the designed Raman system is feasible for in situ spectral acquisition and determining composition in our wind tunnel environment.

5.2 | Influence of air flow and mechanical vibration

The influence of air flow and mechanical vibration on the Raman system was examined by comparing peak intensity of the Raman spectra before and after the start-up of wind tunnel. However, the air flow will heat the sample, and the increased temperature affects the intensity of the Raman peak. Therefore, the measurement point is moved slightly away from the nozzle to ensure that the temperature does not change immediately after the airflow is started, so that the difference before and after the start can be compared in similar temperature.

Significantly, at the moment of the airflow start-up, a large amount of water mist will be blow out from the nozzle in the test wind tunnel. This process lasts about 0.5 s . The water mist dramatically attenuates the detectable Raman signal in one spectrum. Therefore, the plummeting of Raman intensity can be used to mark the start-up of the wind tunnel.

In situ Raman spectra before and after wind tunnel start-up are shown in Figure 7. We can see an unnatural drop in Raman intensity at the sixth second at the start of the wind tunnel. In order to show the changes more

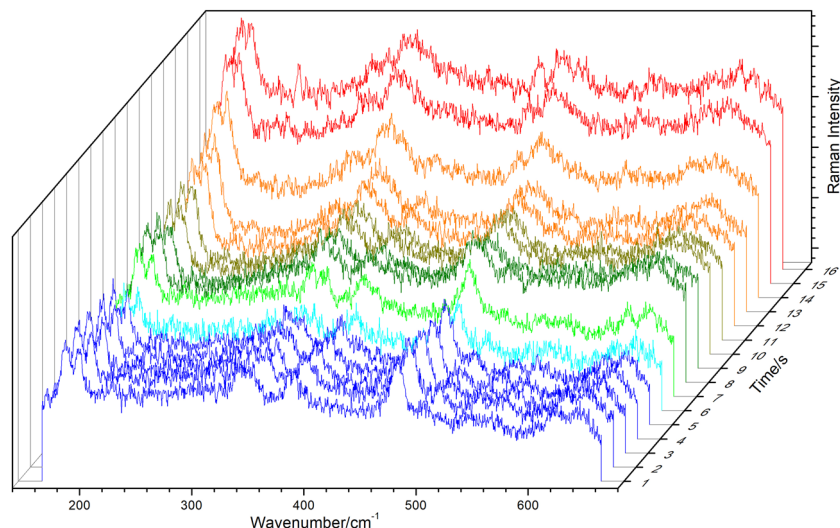


FIGURE 7 In situ Raman spectra before and after wind tunnel start-up [Colour figure can be viewed at wileyonlinelibrary.com]

clearly, the Raman peak at 474 cm^{-1} was fit by Gaussian function and the position and intensity were shown in Figure 8. The Raman peak intensity at H0 decreased significantly, indicating the start of the wind tunnel. RT1 to RT5 represent measurements at room temperature before start-up. From H1 to H10, the peak position at 474 cm^{-1} red shifts about 13 cm^{-1} with the temperature increased. Through the peak position, we can obtain the temperature of the measuring point. The peak position at H1 is the same as that at RT1 to RT5, indicating that the

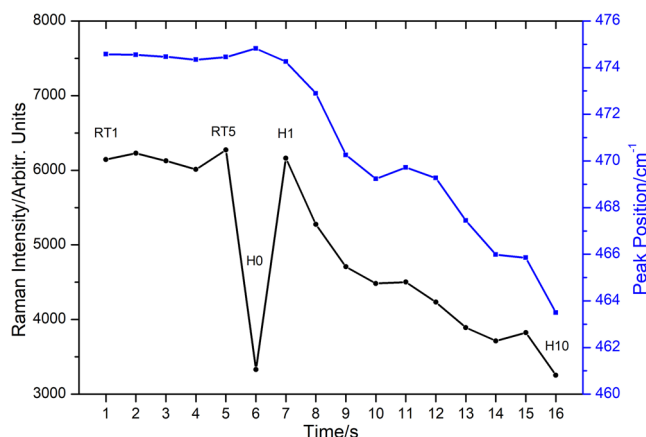


FIGURE 8 The peak position (blue) and peak intensity (black) at 474 cm^{-1} of in situ Raman spectra fit by Gaussian function [Colour figure can be viewed at wileyonlinelibrary.com]

5.3 | Thermal emission background

The Raman spectra with the same intensity scale at different temperature obtained in wind tunnel experiments are shown in Figure 9. It can be clearly assigned to Raman active modes ($176, 186, 332, 370, 466, 625\text{ cm}^{-1}$) of monoclinic zirconia at 1373 K and the Raman active modes ($246, 295, 429, 621\text{ cm}^{-1}$) of tetragonal zirconia at 1773 K . Compared with the Raman spectrum at room temperature, the background does not change significantly with the increase of temperature. At the highest temperature of this experiment, 2073 K , the Raman active modes of tetragonal zirconia can still be identified, and the influence of thermal radiation is limited. It is proved that the high temperature thermal emission background is effectively suppressed.

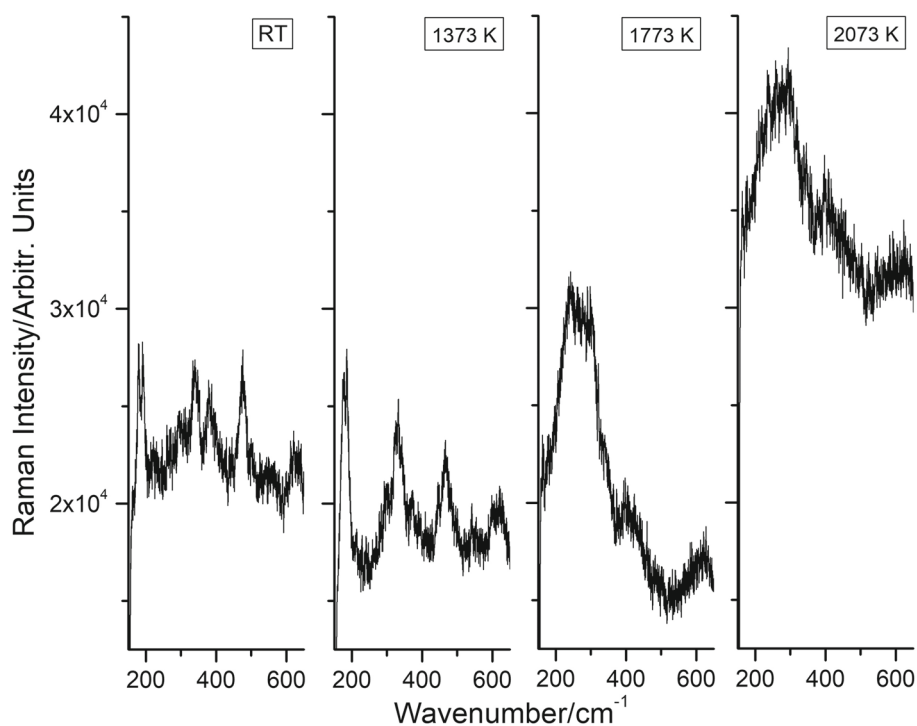


FIGURE 9 The Raman spectra with the same intensity scale at different temperature obtained in wind tunnel experiments

6 | CONCLUSION

The design of an in situ Raman system for hypersonic wind tunnel has been discussed, and the results are reported. Time-resolved method is adopted to reduce the influence of thermal emission. The disturbance of airflow and mechanical vibration is suppressed effectively by a special optical path design with a fiber optic bundle. In situ Raman spectra of zirconia were obtained from room temperature to 2073 K under hypersonic flow. The Raman peak intensity before and after the airflow is compared, which proves that the anti-interference optical path design can eliminate the influence of aero-optical effects and mechanical vibration, and the Raman system can support the in situ quantitative analysis in hypersonic wind tunnel.

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DATA AVAILABILITY STATEMENT

Data available on request from the authors

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